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Robotics | North America

When Factory = Robot

Gen-AI is driving a 'Cambrian explosion' of robots aimed at replacing human tasks in factories. But it's worth considering whether the entire manufacturing process itself must change. What if factories were designed as one giant robot, with human intervention only as an afterthought?



Now may be time for a grounds-up manufacturing rethink - clean slate....

- **In Elon Musk's 5 Step Design Process, The Robots Come in Last.** According to Tesla CEO Elon Musk, automation (while important) is [often a bandage for a poorly designed manufacturing process](#)- cementing critical design while making them occur faster and be much more expensive. Walter Isaacson's 2023 biography of Musk outlines multiple accounts of Musk surveying the Model 3 production line and literally 'ripping out' robots on the spot if he felt they were not 100% necessary. Before considering bringing in the robots: question the requirements, simplify, delete, *delete*... **The best robot may be no robot.**
- **Time to Reflect. Not all manufacturing is created equal.** Has manufacturing been *truly* transformed since Henry Ford's first moving assembly line in 1913? The emergence of GenAI is creating a moment of 'pause' for engineers to re-assess fundamental rules that have been learned and passed down for decades. With all the talk of "rebuilding US manufacturing" - it's incredibly important for policymakers to incentivize the *right kind* of manufacturing.
- **Countering China's Robot & Labor Advantage.** According to the [International Federation of Robotics \(IFR\)](#), in 2024, China accounted for 54% of all industrial robots installed globally. Our conversations with robotics experts indicate that potentially >90% of the critical components supply chain is in some way controlled by China. Comparing the scale of human manufacturing workforces paints a similar story. The Chinese and US manufacturing ecosystems are fundamentally distinct and almost certainly will remain so. *Necessity is the mother of invention.*
- **An Agentic Manufacturing Future.** Imagine a single AI system that managed everything from design, procurement, production, and out-bound logistics - not just one factory, but the entire supply chain as a unified apparatus. Every machine interconnected as one. A bottleneck appears? Everything else autonomously adjusts. Need to build a new line? Designs are prepared and equipment is already on the way - the robots will assemble it. Running thin on inventory? Output is toggled up and material orders are automatically placed with an upstream factory - every node in the supply chain reacts accordingly. Facilities replicate like cells in biology and wither off when not

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needed. *If this sounds overly futuristic, we would highlight that this is not an idea we've dreamt up ourselves. It's becoming clear to us that the concept of rethinking manufacturing as a fully AI-defined, interconnected system is becoming one of the hottest topics among manufacturing experts, hard-tech founders, and investment funds. A growing number of signs are all around us.*

- **Major Private Companies/Startups to Watch.** We are following a number of recently announced or reported startups that fit under this 'factory = robot' thesis. *To name a few:*

- Amazon founder Jeff Bezos is leading a stealth startup called **Project Prometheus**, his first CEO position since 2021, to develop AI for engineering and manufacturing (cars, aerospace, electronics, and more) with \$6.2bn in reported funding as of November 2025 ([NYT](#)).
 - Separately, it's recently been reported by the WSJ that Bezos is in the process of raising **\$100bn for an AI manufacturing fund** which is separate from but related to Project Prometheus (Jeff Bezos has not commented to our knowledge). The fund would reportedly buy up legacy manufacturing companies to integrate them with AI. If completed, it would be the largest single fund raise for an investment fund since the \$100bn launch of SoftBank Vision Fund in 2017.
- Former OpenAI Head of Research, Bob McGrew, is reportedly raising \$70mn at a \$700mn valuation, per WSJ, for a new startup called **Arda** which aims to create a software platform to run autonomous factories.
- Rivian founder and CEO RJ Scaringe recently spun out **Mind Robotics** as an independent company to build generalizable AI models, hardware, and deployment infrastructure for factory automation. It is reportedly valued at \$2.0bn [per the WSJ](#).
- Uber founder Travis Kalanick recently launched **Atoms**, a rebranding and expansion of his post-Uber company City Storage Systems which folds in CloudKitchens (ghost kitchens), Pronto AI (autonomous mining vehicles) and others, and aims to build specialized robotics systems for a wide range of industries.
- **In this note we profile 18 startups relevant to the "Factory = Robot" debate: Arda, Atoms, Bright Machines, Cellares, Divergent, Dyna, Genesis AI, Hadrian, Intrinsic, Matter, Mind Robotics, Nominal, Physical Intelligence, Project Prometheus, Skild AI, Tulip Interfaces, Vention, VulcanForms.**

- **Key Public Market Expressions.**

- Our US Auto Team (led by Andrew Percoco) recently visited **Tesla's (TSLA)** Cybercab production at Giga Texas. The company is attempting a grounds-up rethink of auto production- cutting the process down to no more than 20 steps to eventually produce one car every 10 seconds.
 - See below for full team takeaways: [Tesla Inc: Unlocking the Robotaxi Flywheel \(18 Mar 2026\)](#)
- Our US Software team (including Sanjit Singh) highlights **Palantir's**

(PLTR) Ontology as a key enabler of agentic operations for manufacturers, serving as a live digital twin of the business that unifies siloed data from ERP systems, shop-floor sensors, maintenance records, and other enterprise sources into a single operational model. For manufacturers, that means connecting machines, work orders, technicians, and production workflows in real time, allowing teams to automate processes, improve decision-making, and deploy AI agents with grounded business context, governed actions, and auditability across the factory floor.

- See below for further details: [Palantir's Ontology Creates Digital Twin of the Manufacturing Enterprise](#)
- Our US Semis team (led by Joe Moore) continues to rate **NVIDIA (NVDA)** as his Top Pick. The company's [Omniverse simulation platform](#) allows manufacturers to build physics-based digital twins of factories and run real-time simulations which can materially reduce production planning cycles and support more autonomous decision making on the factory floor.
- Our US Multi-Industry Industrials team (led by Chris Snyder) prefers **Eaton (ETN)** and **Rockwell Automation (ROK)** as their favorite US re-shoring picks. ROK is increasing its exposure to agentic manufacturing through its [FactoryTalk](#) software suite, which integrates AI for production scheduling, predictive maintenance, and quality control. The company has also partnered with **NVIDIA** and **Microsoft** to deploy digital twins and edge AI tools that aim to enable real-time, autonomous production adjustments across factory systems.
- Western contract manufacturers are potential key beneficiaries in a US agentic manufacturing re-shoring cycle: **Jabil (JBL)**, **Flex (FLEX)**, **Celestica (CLS)**, **Sanmina (SANM)**, **Plexus (PLXS)**, **Benchmark Electronics (BHE)**. *Not covered by MS Research.*

Exhibit 1: Key Public & Private Companies for the 'Factory = Robot' Theme

Manufacturing as a Service		Factory Brain		General-Purpose Foundation Models	
Public		Public		Public	
FOXCONN HON HAI TECHNOLOGY GROUP	PEGATRON	Alphabet intrinsic	SAP	Google DeepMind	Meta
金寶集團 KINPO GROUP	JABIL	C3.ai	Palantir	TESLA	amazon
PLEXUS	flex	ptc	YASKAWA	Alibaba.com	Tencent
Celestica	Benchmark	HEXAGON	Rockwell Automation		
Private		Private		Private	
Bright Machines	CELLARES	Honeywell	Schneider Electric	SKILD AI	Genesis AI
DIVERGENT	HADRIAN	VENTION	TULIP	Physical Intelligence (π)	covariant
matter	VulcanForms	NOMINAL	COGNITE	Generalist	General Robotics
		BRAIN CUBE	ODEN TECHNOLOGIES	FieldAI	DYN A
		MIND ROBOTICS	LITMUS		

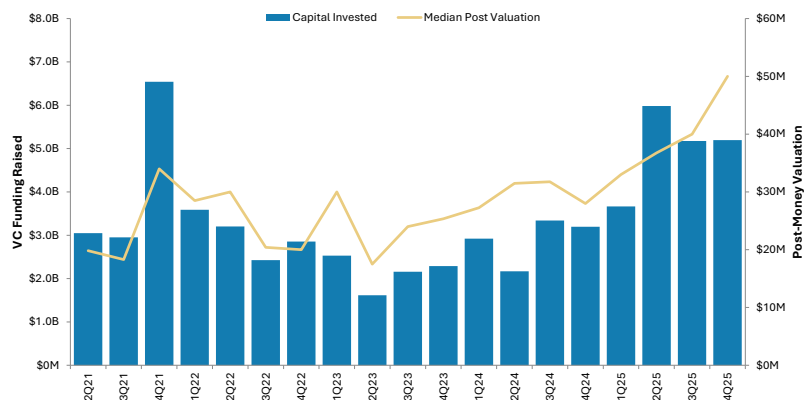
Source: Company Websites, Morgan Stanley Research

Exhibit 2: Elon Musk's 5 Step Design Process

Elon Musk's 5-Step Design Process

The Algorithm for Radical Efficiency & Innovation

- 1 Make the Requirements Less Dumb**
Question every requirement. Be especially skeptical of "smart" requirements from intelligent people.
- 2 Delete the Part or Process**
Remove unnecessary components. If you're not adding things back at least 10% of the time, you're not deleting enough.
- 3 Simplify and Optimize**
The most common error of a smart engineer is to optimize a thing that should not exist. Only optimize after deletion.
- 4 Accelerate Cycle Time**
Speed up the process. Compress timelines and iterate faster. Make the process as quick as possible.
- 5 Automate**
Only after completing the first four steps. Automating a bad process makes the mistakes faster and more expensive.

Key Insight: The order matters. Never skip to automation first!Source: Morgan Stanley Research, Various interviews from Walter Isaacson's *Elon Musk* (2023)**Exhibit 3:** US Venture Funding for Manufacturing-Related Startups Has Been Steadily Increasing Since 2Q23.

Source: PitchBook, Morgan Stanley Research

For Further Reading:

- [Rowdy Robot: Physical AI: America's Chance to Re-Invent Manufacturing \(10 Feb 2026\)](#)
- [Physical AI: The Morgan Stanley Robot Almanac \(7 Dec 2025\)](#)
- [US Multi-Industry: US Reshoring Flame Has Been Lit For \\$10tn Opportunity \(16 Oct 2024\)](#)
- [Space: Varda Space: Space Manufacturing, Pharma Made in Orbit \(19 Feb 2026\)](#)
- [Robotics: Agile Robots: Das Bot \(17 Mar 2026\)](#)
- [Robotics: VulcanForms: Metals Manufacturing, Made Simpler \(5 Feb 2026\)](#)
- [Embodied AI: World Models: AI's Journey from Digital to Physical \(22 Mar 2026\)](#)
- [Rowdy Robot: OpenAI RFP Aligns with Rebuilding American Manufacturing Muscle \(22 Jan 2026\)](#)

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The Agentic Manufacturing Thesis

Could Now be the Time for a Grounds-Up Rethink? Has manufacturing been *truly* transformed since the Ford assembly line in 1913? The advent of AI is creating a moment of 'pause' for engineers to re-assess fundamental rules that have been learned and passed down for decades. At the same time, US manufacturing as a % of GDP has steadily declined from ~30% of GDP in the 1950's to <10% today. Manufacturing as a % of non-farm payrolls paints a similar picture - steadily declining to <8% today.

Greenfields Galore? As US policy-makers continue to assess the risk posed by the growing reliance on foreign manufacturing, our US Multi-Industry team (led by Chris Snyder) *continues to believe* the US is entering the early innings of re-Industrialization, a multi-decade opportunity sized at ~\$10tn that could restore growth to the US Industrial economy. *Why rebuild with last-gen manufacturing?*

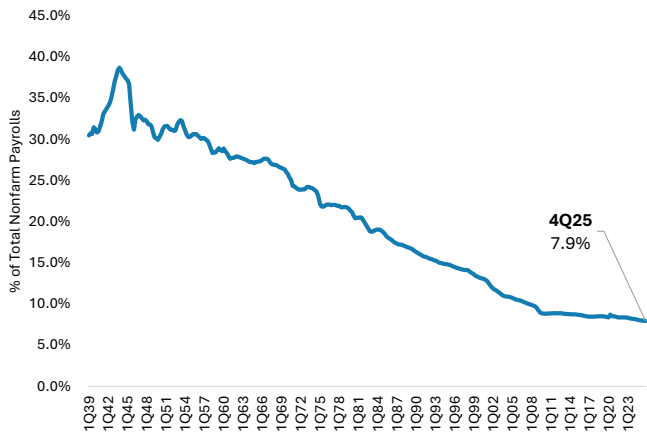
Dark Factories a Red Herring? Robots have been building robots for decades. *In 2003, CNN documented* the impressive lights-out manufacturing (AKA a 'dark factory') of Japanese robot OEM Fanuc where robot arms were automatically assembled at a rate of 50 per day - no lights, AC, heat... *or AI*. Very impressive. So why isn't all manufacturing today 'lights out'? Based on our conversations with manufacturing experts, the traditional dark factory model (while impressive) can be incredibly rigid with potentially hundreds of closed-loop specialized systems that still need to be individually managed. Great if you're a *world-class electronics manufacturer*, but otherwise prepare for a lot of precision engineering, redesigns, and debugging.

Manufacturing 2040: Factory Hivemind. Imagine a single AI system that managed everything from design, procurement, production, and out-bound logistics - not just one factory, but the entire supply chain. Every machine interconnected as one. A bottleneck appears? Everything else autonomously adjusts. Need to build a new line? Designs are ready and equipment is already on the way - the robots will assemble it. Running thin on inventory? Output is toggled up and material orders are automatically placed with an upstream factory - every node in the supply chain reacts accordingly. Facilities replicate like cells in biology and wither off when not needed.

Manufacturing as a Service (MaaS) 2.0. In this envisioned future, AI-driven networks could enable companies to tap into and dynamically route demand to the necessary production capacity. Demand signals trigger autonomous build-outs, supply chains self-configure, and capacity scales up or down without human intervention.

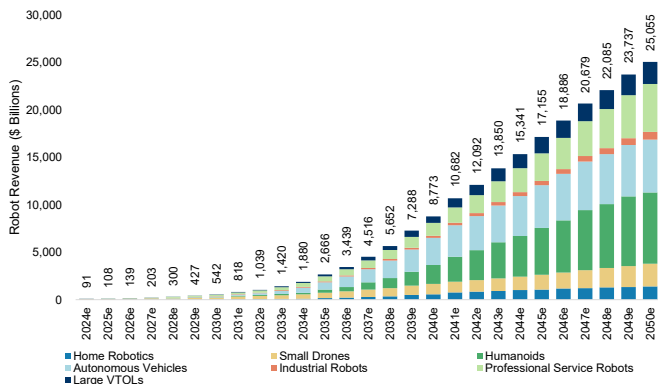
On Humanoids. If the factories aren't designed for humans - why build robots that are designed to mimic every feature of human morphology? *To be clear, we are still long-term humanoid bulls*, but overtime the definition of a 'humanoid' especially in factory settings may blur. There will be many instances where a general purpose robot with human or above-human dexterity play vital roles in this potential autonomous manufacturing future.

Exhibit 4: Manufacturing as a % of Total US Nonfarm Payrolls



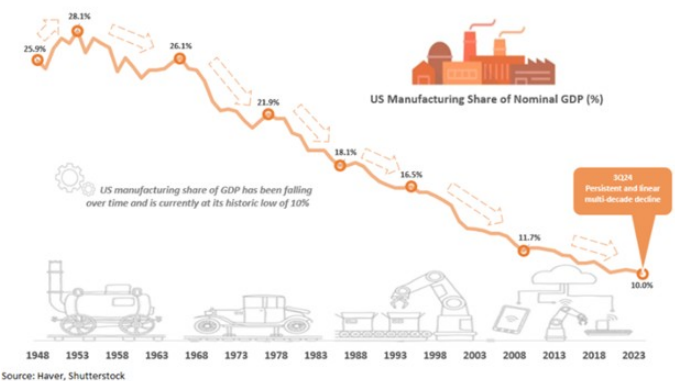
Source: US Bureau of Labor Statistics, Morgan Stanley Research

Exhibit 6: Morgan Stanley Global Robot Revenue Projections (Reach out for Full Model)



Source: Morgan Stanley Research Estimates

Exhibit 5: Manufacturing as a % of Nominal GDP



Source: BEA, Haver, Shutterstock, Morgan Stanley Research

Exhibit 7: Dark Factories: Impressive feats, but Pre-AI



Source: Xiaomi

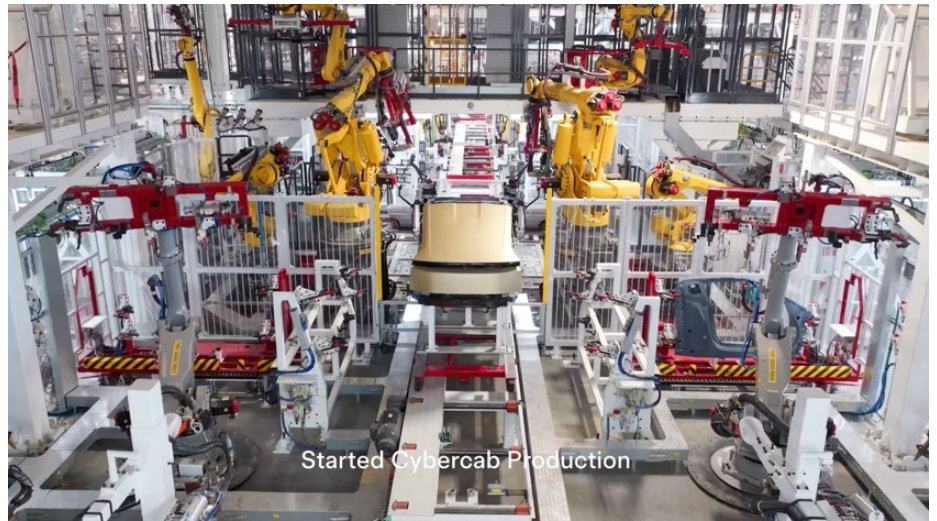
Tesla Cybercab's Rethink of Auto Manufacturing

For more details on the US Auto Team's (led by Andrew Percoco) recent visit of Giga Texas, please see: [Tesla Inc: Unlocking the Robotaxi Flywheel \(18 Mar 2026\)](#)

Cybercab - Tesla Ushers in Auto Manufacturing of the Future: Tesla is making several ground-breaking changes to the traditional auto manufacturing process with its Cybercab production line in Austin. **The manufacturing process uses a modular “unboxed” architecture that replaces traditional body in white, paint, and sequential assembly.** The vehicle is built from five major modules – front, central battery, rear cargo, and two side modules – that are assembled in parallel and later combined by a framing system. All exterior panels are plastic and produced via reaction injection molding, with color injected directly into the plastic. This approach allows greater access during assembly and reduces factory footprint, largely due to eliminating the paint shop. Through this process, Tesla aims to meaningfully reduce the robotaxi cost structure to ~\$0.30/mile, which is well below other robotaxi and rideshare peers. Start of production is still on track for April 2026.

- **A typical automotive plant has hundreds of discrete production steps** following material flow into a plant (ie steel coil) into stamping, metal-forming, body shop/ welding, engine/transmission machining, paint shop, powertrain marriage, wiring harness, trim installation, final assembly.
- Our understanding is that manufacturing of Cybercab is achieved **in no more than 20 steps** in its novel ‘unboxed’ assembly process where the car is made from the inside out.
- Cybercab **does not have a steering wheel**, pedals, mirrors and many other components that would be required for a human operated vehicle.
- Cybercab has only 2 seats and a highly spartan interior (just 1 screen).
- Cybercab's **structural frame is made from 3 pieces (front, rear and center)** in a 6,500 to 9,000 ton Idra gigapress. This process replaces hundreds of parts and thousands of welds vs. a typical car body shop.
- Cybercab **does not have a traditional paint shop**. The exterior of the car is made from polyurethane body panels where the color is injected directly when the panels are molded.
- Cybercab uses an advanced Etherloop (Ethernet-based) architecture with integrated 48V power and data transmission, significantly reducing the number of traditional **wiring harnesses to ~5–7 modular bundles vs. 40 or more bundles in a traditional car**, saving multiple kilometers of total wiring length and copper content.
- Tesla targets a **production cycle time of 10 seconds** per Cybercab which compares to 60 to 90 seconds for a high volume conventional human-driven car.

Exhibit 8: Tesla Cybercab Manufacturing



Source: Tesla

Palantir's Ontology Creates Digital Twin of the Manufacturing Enterprise

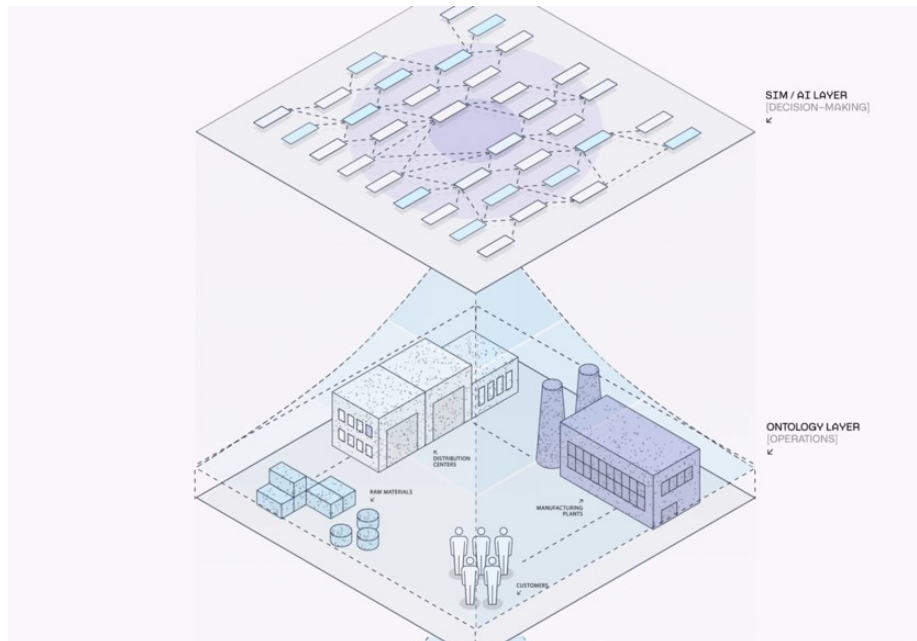
For more details on the US Software team's view on Palantir's Ontology as a unique and durable moat see: [Ontology - What Is It and How Durable of an Advantage Does It Represent?](#)

Palantir (PLTR) is covered by Sanjit Singh

Palantir's Ontology is best understood as a live digital twin of an enterprise: it takes siloed data from systems such as ERP, CRM, databases, sensors, and operational documents, and maps that data to the real-world entities, relationships, and actions that define how a business actually runs. In practical terms, Ontology creates a shared operational model built around "objects" (for example, machines, work orders, suppliers, technicians, or production lines), their attributes, the links between them, and the governed actions that users or software agents can take. Because those core elements are tied directly to continuously updated source systems, the **Ontology is not a static dashboard layer but a real-time representation of the business that makes it easier to build applications, automate workflows, and deploy AI agents** on top of the same common data foundation.

For manufacturers, the value proposition is especially compelling because manufacturing environments are typically asset-heavy, operationally complex, and fragmented across many systems.

- Ontology can connect machines, maintenance records, work orders, inventory, suppliers, technicians, plant metrics, and scheduling systems into one governed model, replacing manual cross-system reconciliation with a single operational view.
- That allows plant and operations teams to monitor performance in real time, identify underperforming equipment, coordinate maintenance, improve production decisions, and reduce "swivel-chair" workflows across functions.
- Just as importantly, **Ontology provides the context layer that lets AI applications and agents act on live operational data** rather than stale training sets, with defined action boundaries, write-backs into source systems, and full audit trails.
- In that sense, Ontology is not just a data integration layer for manufacturers; **it is an operational system that can help unify the factory floor and enterprise back end**, making production environments more automated, more intelligent, and more responsive over time.

Exhibit 9: Ontology Layer Representing a Manufacturing Campus

Source: Palantir

US Reshoring \$10tn Opportunity

For more from our US Multi-Industry Industrials Team (led by Chris Snyder), please see:

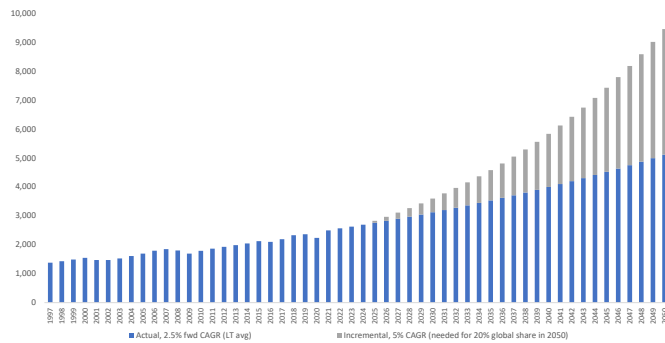
- [Multi-Industry: 2026 Outlook: Relative Stability Layers in Cyclical Torque \(6 Jan 2026\)](#)
- [US Multi-Industry: US Reshoring Flame Has Been Lit For \\$10tn Opportunity \(16 Oct 2024\)](#)

We believe the US is entering the early innings of re-Industrialization, a multi-decade opportunity that we size at \$10tn and believe will restore growth to the US Industrial economy following 20+ years of stagnation. The US Reshoring theme aligns closely with MS Thematics' [Investing for a Multipolar World](#) – underpinning our thesis are structural tech advancement and sharper focus on operational resiliency after the pandemic and subsequent supply chain disruption. Since China joined the World Trade Organization (WTO) in 2000, the US has lost 9 PP of Global Manufacturing share (~\$1.5tn annual output), flipping US trade from balanced to a \$1tn annual deficit and driving declines in US Manufacturing fixed asset investment. But the tide is turning – since 2020, the US is attracting foreign direct investment (FDI) at a rate not seen since the 1990s and US Manufacturing construction has surged ~300% with project starts stabilizing at ~3x pre-Covid levels. After decades of chasing low cost labor, we believe that access to capital and proximity to end users will now drive the regional allocation of capex - parameters to which the US has no equal. While Reshoring is broadly positive for all US Industrials given the regional exposure, **ETN and ROK** are our preferred stocks on the theme.

The US Reshoring flame has been lit and momentum continues to build. The flame was first lit in 2018 as Trump tariffs kickstarted boardroom conversations around global supply chains for the first time in 40 years and brought noticeable change. Covid then brought kerosene to the flame with historic economic disruption due in large part to disrupted trade as the true cost of long rigid supply chains was revealed. The disruption and reduced confidence in key trade partners led domestic policy centered above all else on padding the resilience of the domestic economy and reducing the risk of systemic shocks – further fanning the flames. Now the torch has been passed to Tech diffusion, a structural tailwind for US Manufacturing where latest genAI advancement both brings down the US Manufacturing cost curve while fueling the next round of global protectionism. The opportunity is compounded by 25 years of US Manufacturing under-investment, leaving **billion dollar companies going after trillions of opportunity.**

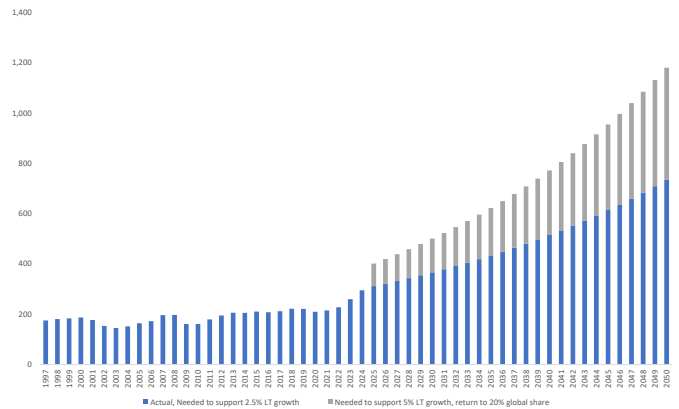
Following two decades of under-investment and stagnation, we believe the US is in the very early innings of re-industrialization - a multi decade opportunity which we size at over \$10T (incremental IP + Capex). Our US Reshoring thesis is underpinned by structural tech advancement and heightened focus on operational resiliency following the Covid pandemic and subsequent supply chain disruption. Gone are the days where Global Mfg chased low cost labor; moving forward, we believe capital and proximity to the end user will dictate the regional allocation of capex - parameters to which the US has no equal.

Exhibit 10: US Reshoring: Opportunity for ~\$5T incremental US Mfg Output (\$bil, real)



Source: US Census Bureau, Morgan Stanley Research.

Exhibit 11: US Reshoring: Opportunity for ~\$6T incremental US Mfg Capex (\$bil, real)



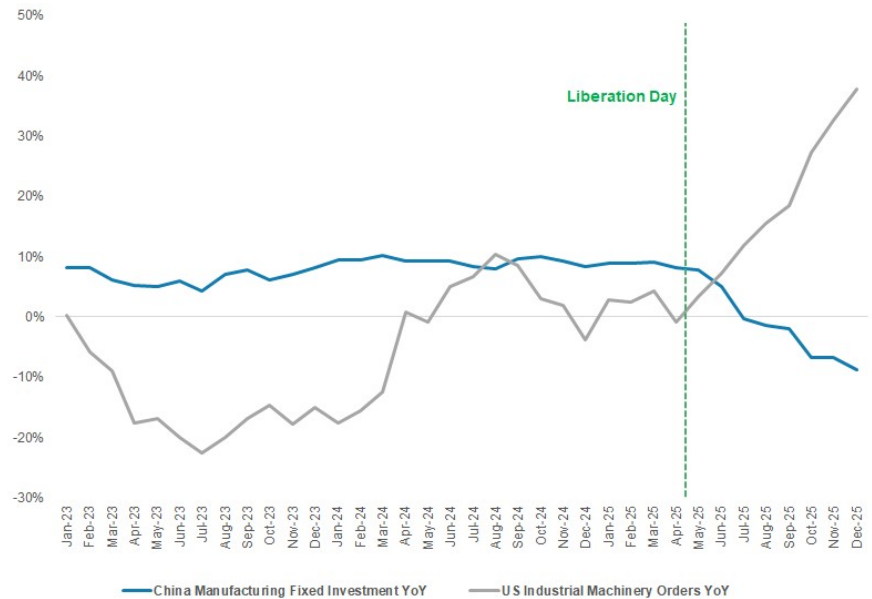
Source: US Census Bureau, Morgan Stanley Research.

Exhibit 12: US Industrial Machinery Orders (Annual, \$mil, Real): Down ~50% from 2000 to 2025 post China WTO inclusion, highlighting the opportunity at hand from our \$10T US Reshoring thesis



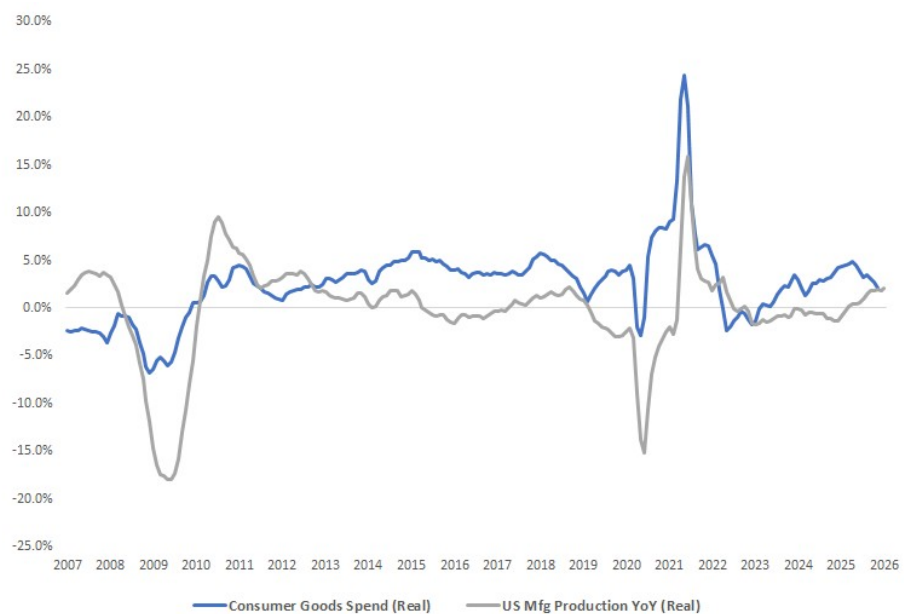
Source: US Census Bureau, Morgan Stanley Research

Exhibit 13: We believe that due to US purchasing power (~30% global consumption), companies must serve the US market and 'Liberation Day' tariffs changed the cost calculus on landing goods into the country. We have since seen the incremental unit of investment shift towards the US market, which signals where the next Greenfield facility lands when it is needed, all in line w/ our \$10T US Reshoring thesis



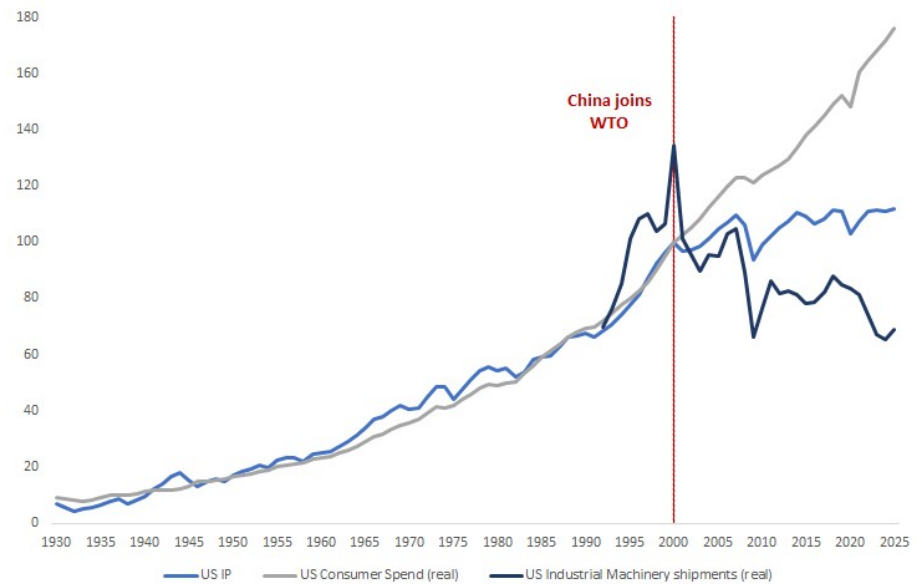
Source: US Census Bureau, Morgan Stanley Research

Exhibit 14: US Consumer spend has consistently & materially outpaced US Mfg production over the last 20yrs due to outsourcing, but the delta evaporated to exit 2025 – signaling both US Reshoring & channel destock, a positive for the cycle into 2026



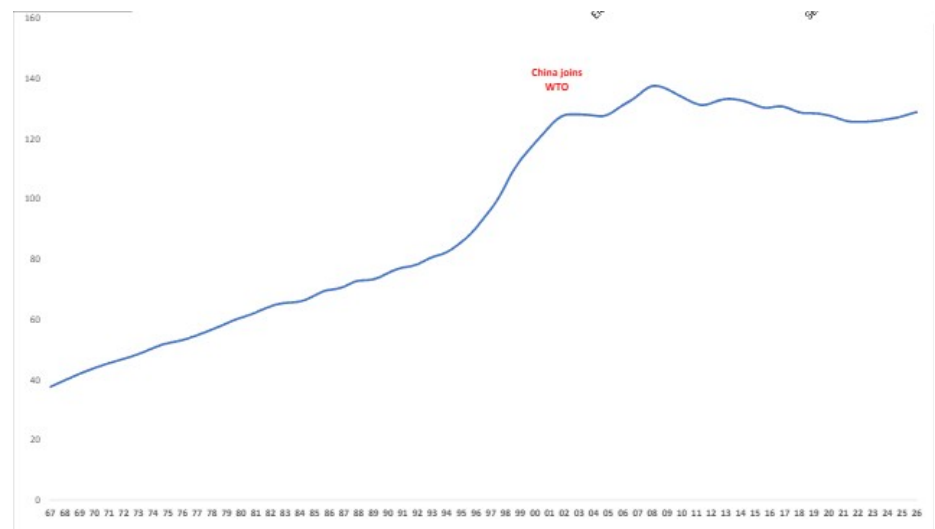
Source: US Census Bureau, Morgan Stanley Research

Exhibit 15: While US Industrial Production has tracked sideways since China entered the WTO, US Manufacturing Capex has fared even worse w/ US Industrial Machinery shipments down ~50% in real terms - setting the stage for a strong equipment upgrade cycle as US policy + structural tech diffusion tilt the scales back towards domestic production



Source: US Census Bureau, Morgan Stanley Research

Exhibit 16: Total US Manufacturing capacity (Indexed) - realizing material growth for the first time in over a decade



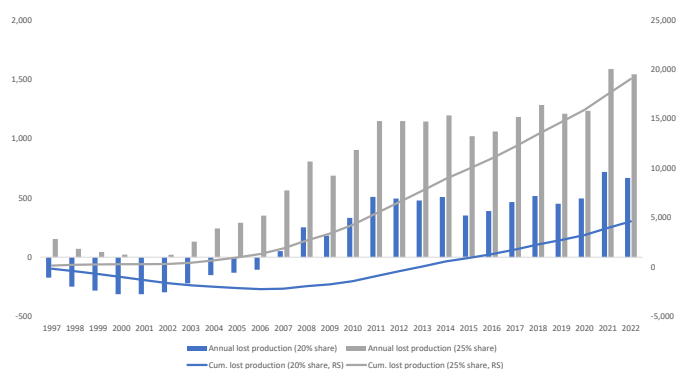
Source: US Census Bureau, Morgan Stanley Research

Quantifying what was lost

If the US was able to sustain the 25% Global Mfg share it commanded in the late 1990s, annual lost output would be currently tracking at \$1.5T pa or \$19T cumulative since China joined the WTO. However, we view 20% Global share as a more sustainable level and aspirational target moving forward – a benchmark that would imply the US has lost \$700B annual output (\$6T cum). Our math suggests US Mfg Fixed asset base would need

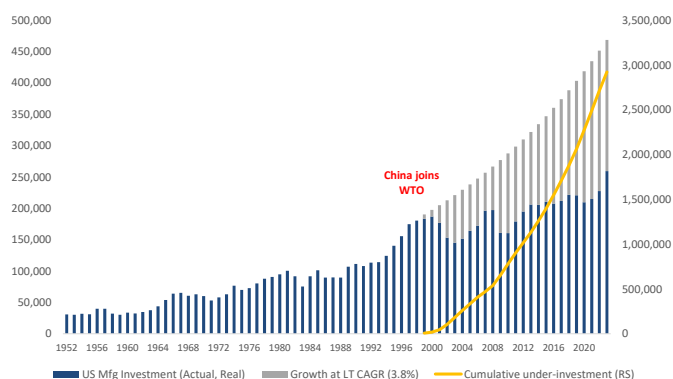
to increase by more than \$4T in real terms to support a return to 20% Global Mfg share by 2050, an incremental \$3.5T above the trailing 25yr trend line. This would require US Mfg Capex to grow at a 4.4% CAGR over the next 25yrs, more than tripling the annual spend to \$1T+. While this ~4% forward CAGR comes in nearly ~10x above levels observed from 2000-2022, it only matches the growth realized during the prior four decades, before China joined the WTO.

Exhibit 17: US Mfg lost output at 20% and 25% share (\$bil)



Source: US Census Bureau, Morgan Stanley Research.

Exhibit 18: US Industrial Economy has been under-invested in by \$3T since China joined the WTO in 2000 (\$mil)

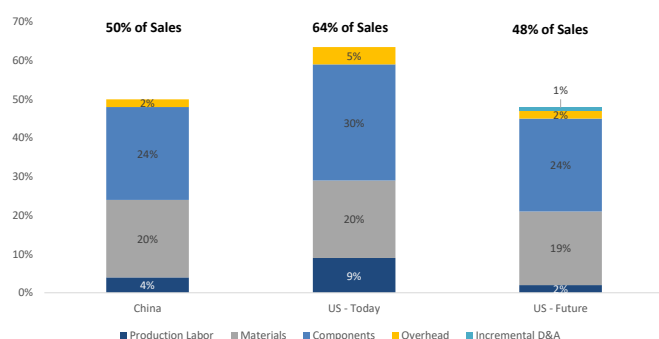


Source: US Census Bureau, Morgan Stanley Research.

Gotta Make Dollars to Make Sense

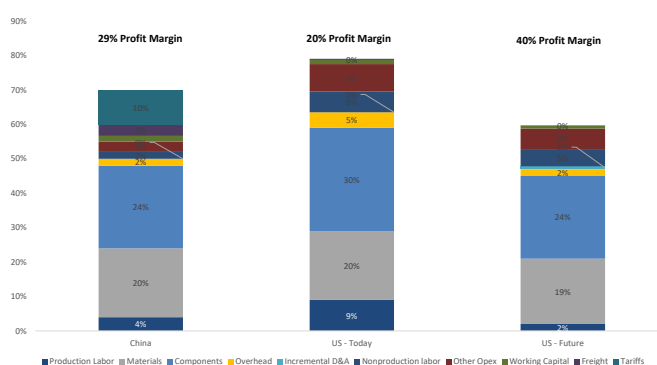
For US Reshoring to prove structural, the decision must make sense at a micro economics level. While our detailed Mfg cost benchmarking analysis pegs US output ~25% above China on a purely production basis, we see a much more narrow high single digit delta when applying a more sophisticated total cost approach. We believe a comparable cost structure today combined with clear and increasingly important indirect benefits is enough to drive US Reshoring today. Perhaps most importantly, companies that are Reshoring today can take confidence in the fact that Technology advancement only goes one way and will allow the costs of US Mfg to continue to narrow vs low-cost International trade partners.

Exhibit 19: Direct Production Costs as % of Sales: US tracking +25% vs China



Source: US Census Bureau, China Nat'l Bureau of Statistics, Morgan Stanley Research.

Exhibit 20: Total Cost Approach: US tracking just +HSD vs China



Source: US Census Bureau, China Nat'l Bureau of Statistics, Morgan Stanley Research.

Public Beneficiaries: Western Contract Manufacturers

We believe North American contract manufacturers are a highly-relevant way for investors to gain exposure to the US reshoring + physical AI theme. A number of these companies have already announced capacity expansions in North America including Jabil's pledged expansion of US facilities to support cloud and AI hardware, Flex's build-out of advanced manufacturing capacity in Texas, Celestica's scaling of North American production for AI servers, and Sanmina's announced investments in domestic energy manufacturing. At the same time, several EMS players (see [JBL](#), [FLEX](#)) are integrating AI-enabled manufacturing platforms such as digital twins, real-time production analytics, and automated scheduling tools that aim to improve yield, reduce downtime, and allow production lines to adjust to demand changes with limited human input, which aligns with the shift toward agentic manufacturing systems.

What are contract manufacturers/electronic manufacturing services (EMS) companies? Contract manufacturers are companies that manufacture products for other brands, handling the production work while the brand focuses on design and sales without having to own factories or worry about the complexities of manufacturing. EMS is a subset of contract suppliers that produce electronic products across a range of range of industries from consumer electronics, AI, energy, and industrials. Most of these companies handle printed circuit board assembly (PCBA) - essentially placing components (chips, connectors, etc.) onto boards, as well as handling the testing and final assembly build.

The world's largest are Taiwanese companies [Hon Hai Precision \(AKA 'Foxconn'\)](#) and [Pegatron](#) which collectively brought in nearly \$300bn in revenue in 2025 - and are known for being key suppliers for major global electronics companies such as Apple, Microsoft, NVIDIA, Sony and others.

EMS is most known for consumer electronics, but it also extends to key sectors such as aerospace, industrial equipment, and robotics, where specialized suppliers build key components and systems that support infrastructure, defense capabilities, and advanced manufacturing. ***To put it simply, if you want to produce any physical AI device (from robots to consumer devices to drones) you need large quantities of assembled printed circuit boards (PCB's).***

Contract Manufacturers to Know:

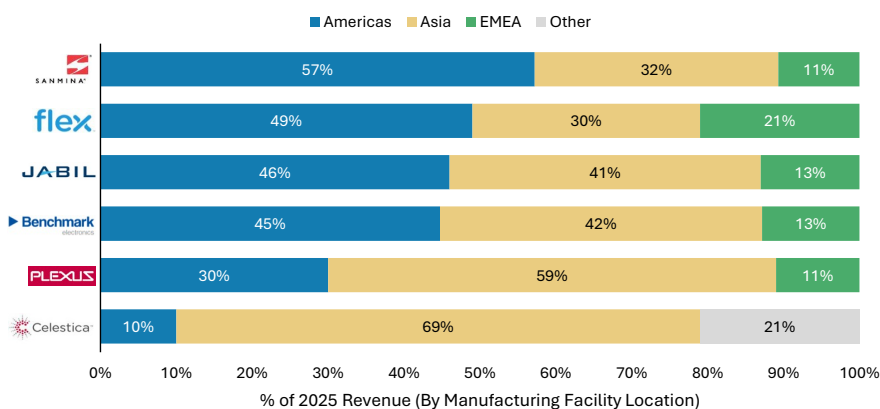
- **Flex (FLEX):** [Flex](#) is a global electronics manufacturer that serves customers across cloud infrastructure, automotive, healthcare, and industrial markets, with a large footprint across the US, Mexico, and Asia. The company differentiates through its scale and power electronics capabilities, including [acquisitions](#) in power distribution and [liquid cooling](#) that target AI data center infrastructure. It has been expanding North American capacity, including facilities in [Texas](#) and [Mexico](#), supported by a [footprint](#) of ~7m sq ft in the US and ~9m sq ft in Mexico.
- **Jabil (JBL):** [Jabil](#) is a US-based contract manufacturer that serves cloud, capital equipment, automotive, and healthcare customers, with more than 30 manufacturing sites in the US. It stands out for its exposure to hyperscaler and data center customers including [Amazon](#). It also holds a [partnership with US humanoid startup Apptronik](#) to manufacture and deploy their 'Apollo' humanoids.

The company has committed ~\$500m to expand US manufacturing for AI data center hardware, with a new facility expected to come online by mid-2026.

- **Celestica (CLS):** Celestica is a Canada-based manufacturer (and former IBM manufacturing unit) that produces AI servers, networking equipment, and storage systems for large cloud and technology companies. The company has been heavily focused on AI infrastructure opportunities, with a recent partnership with AMD and strong growth (28.5% y/y in 2025) tied to hyperscaler demand. It has been expanding North American capacity, including investments in Texas to support AI hardware production and data center buildouts.
- **Sanmina (SANM):** Sanmina is a US-based electronics contract manufacturer that serves customers in communications, cloud & AI infrastructure, industrial, medical, and A&D markets. Uniquely, the company also has extensive in-house PCB fabrication capabilities (benefits vertical integration but also more capital intense) compared to the other suppliers which mostly source bare PCB's from dedicated manufacturers in Asia. Late last year, the company announced a major expansion of its energy business with a new factory in Houston to produce transformers and other energy-infrastructure products.
- **Benchmark Electronics (BHE):** Benchmark is a US-based manufacturer focused on lower-volume, high-complexity products for aerospace, defense, medical, and industrial customers. The company differentiates through engineering-intensive programs and precision manufacturing, which are less exposed to consumer electronics cycles.
- **Plexus (PLXS):** Plexus is a US-based manufacturer that provides design, engineering, and production services for healthcare/life sciences, aerospace and defense, and industrial customers. The company focuses specifically on high-complexity products in demanding regulatory markets.

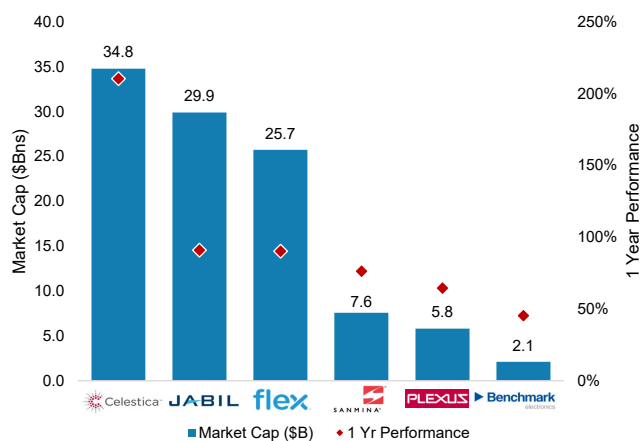
These stocks are not covered by Morgan Stanley Research and Morgan Stanley is not making an implicit recommendation for these companies. We include this text solely to indicate that these companies are exposed to the theme of physical AI and domestic onshoring.

Exhibit 21: 2025 Revenue by Manufacturing Location

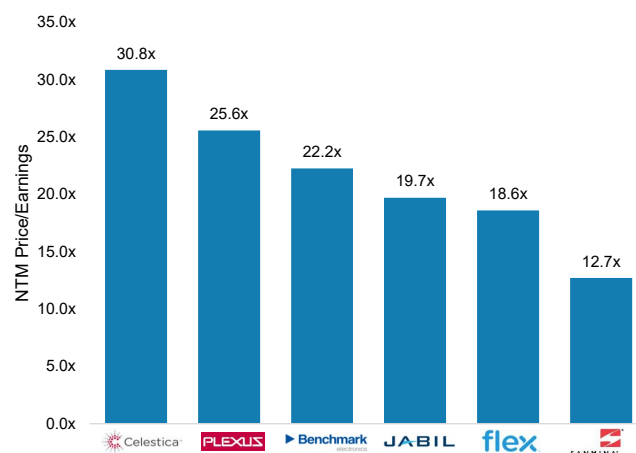


Note: CLS only reports revenue for countries with >10% of revenue. Thailand and Malaysia represent a combined 69%. Americas ≠ United States. For example, SANM indicates that ~67% of North America revenue is Mexico and ~30% United States.

Source: Company Data, Morgan Stanley Research

Exhibit 22: Market Cap & 1 Year Performance

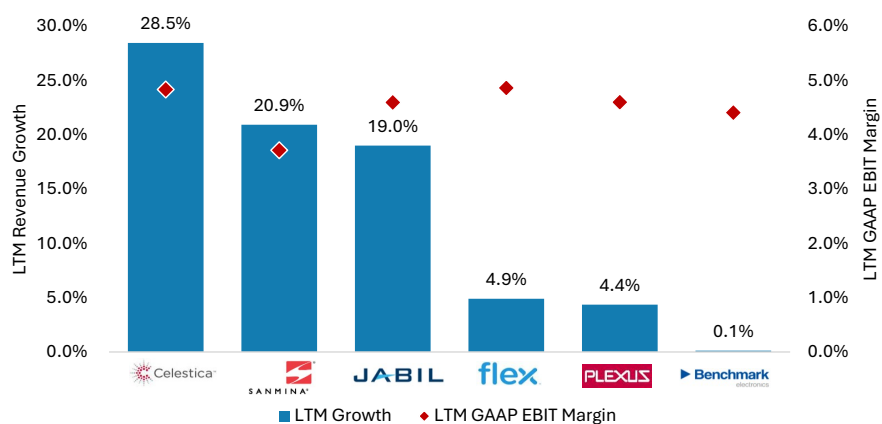
Source: FactSet, Morgan Stanley Research

Exhibit 23:a NTM Price/Earnings

Source: FactSet, Morgan Stanley Research

Exhibit 24:

LTM Revenue Growth and GAAP EBIT Margins



Source: FactSet, Morgan Stanley Research

Startups to Follow: Manufacturing as a Service

These are companies that design, automate, and operate entire factories, delivering finished products using AI-driven production systems. They either operate a contract manufacturing/manufacturing as a service (MaaS) model where revenue is generated based on goods produced or sell/license full-stack manufacturing platforms to customers who maintain and operate the equipment on their end.

Bright Machines

Bright Machines is a US-based software and robotics manufacturing startup developing full-stack 'microfactories' for AI infrastructure assembly. It was founded in 2018 in San Francisco by industry veterans Lior Susan and Amar Hanspal following their work in electronics and manufacturing, with the goal of applying software-centric design and data to factory operations. The company builds modular robotic workcells called **Microfactories** and the Brightware Platform OS to link computer vision, machine learning and production data for automated assembly, inspection and process control in electronics assembly. [Use cases](#) include GPU integration, CPU & heatsink placement, storage systems assembly, and connector handling, among many others to assemble critical AI infrastructure.

Bright Machines integrates AI-enabled software with physical robotics to bring data visibility, traceability and adaptability to factory floors that have historically relied on labor-intensive manual assembly and disjointed automation. Notable partners include Microsoft, Dell, Nvidia, Lenovo, OpenAI, and AWS. The company also claims to have deployed 130+ microfactory systems for 60+ customers across 10+ countries helping to deploy 200k+ servers.

According to PitchBook, the company has raised \$502mn to-date (valuation unknown). Notable investors include Eclipse Ventures, Lux Capital, Blackrock, Nvidia, Jabil, BAM Elevate, and others.

Exhibit 25: Bright Machines (Full-Stack Data Center Infrastructure Assembly)

Bright Machines
San Francisco, CA

Microfactory **Software AI**

Partners

Microsoft, DELL, NVIDIA, Lenovo, OpenAI, AWS

Notable Team Members

Lior Susan
Co-Founder, Chairman
Firms/Others:
Eclipse Ventures
Lahli
Farm 2000

Sviat Dulanov
CEO
Firms:
SPAR
Bain & Company

Fiaz Mohamed
President & Chief Growth Officer
Firms:
Cruise
AWS

Notable Investors
\$502mn Funding Raised (Valuation Unknown)

ECLIPSE, LU+, NVIDIA, Shinhan, BlackRock, JABIL, Microsoft, Balyasny

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

Cellares

Cellares is a biotechnology company developing a fully integrated manufacturing platform for next-generation cell therapies to make drugs more scalable and accessible. It was founded in 2019 by Fabian Gerlinghaus, an aerospace engineer and former Chief Innovation Officer at Synthego who helped scale its CRISPR product portfolio and co-invented the Synthego's RNA synthesizer technology, alongside Omar Kurdi, a bioprocessing expert with deep experience in cell therapy manufacturing and process development. It is based in South San Francisco, CA.

Cellares is attempting to rethink how cell therapies are made. Instead of relying on manual, fragmented lab processes, it has developed a fully integrated, automated platform called the 'Cell Shuttle', which combines multiple steps of production into a single, high-throughput system. The company operates under what it calls an **Integrated Development and Manufacturing Organization (IDMO)** model, pairing the 'Cell Shuttle' platform with large-scale, centralized manufacturing facilities designed for high-throughput production of patient-specific therapies. These 'smart factories' combine the Cell Shuttle systems for manufacturing and a quality control system called the 'Cell Q' for in-line quality control, alongside interconnected software systems that manage workflows and generate electronic batch records. The aim is to enable higher throughput, improved consistency, and reduced cost per batch while supporting a wide range of therapy types. The company claims its model can deliver **up to 10x greater output compared to traditional cell therapy manufacturing approaches**. Its broader goal is to help address one of the biggest bottlenecks in modern medicine by turning a slow, expensive, and labor-intensive manufacturing process into a more scalable, repeatable system that can expand access to life-saving treatments.

According to PitchBook, the company has raised \$612mn to-date from notable investors including 8VC, Eclipse Ventures, Willet Advisors, DFJ Growth, Bristol Meyer Squibb, and Baillie Gifford, among others.

Exhibit 26: Cellares (Fully Automated Cell Therapy Manufacturing)

Source: Company Websites, PitchBook, Morgan Stanley Research

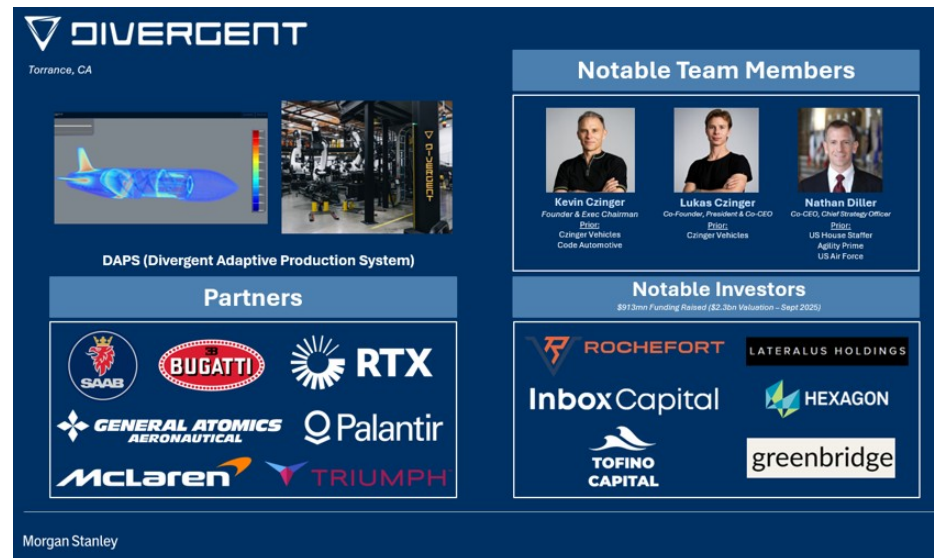
Divergent

Divergent is a US-based manufacturing technology startup developing an end-to-end design and additive manufacturing system for complex vehicle structures. Founded in 2014 in Torrance, California by Kevin Czingier and his son, Lukas Czingier (who both also founded and continue to lead **Czingier Vehicles** - a startup hypercar company), the company set out to develop an end-to-end system for design, additive manufacturing, and robotic assembly of complex vehicle parts. The 'Divergent Adaptive Production System' is a digital platform that combines computational design, additive manufacturing, and automated assembly to produce parts originally for automotive applications, but later expanded to aerospace, defense, and industrial applications. Announced partners to-date include Saab, RTX/Raytheon, McLaren, Triumph Group, and General Atomics, among others.

DAPS integrates generative engineering software, metal additive manufacturing, and robotic assembly into a single digital workflow. Engineers use algorithmic design tools to create lightweight structures, often consolidating multiple components into fewer nodes and structural elements, which the system produces using processes such as laser powder bed fusion. Divergent states that this approach can materially reduce structure weight and lower tooling costs by eliminating the need for stamping dies and molds, while robotic cells handle final assembly through software-defined, fixtureless processes.

According to PitchBook, the company has raised \$913mn to-date from investors including Rochefort Management, Lateralus Holdings, Inbox Capital, Hexagon, and others. It was last valued at \$2.3bn in a \$290mn **Series E round** in September 2025.

Exhibit 27: Divergent (End-to-End Additive Manufacturing for Complex Vehicle Structures)



Source: Company Websites, PitchBook, Morgan Stanley Research

Hadrian

Hadrian is a US-based startup building end-to-end automated machining/

manufacturing facilities supplying the A&D industry. Founded in 2020 in Torrance, CA by entrepreneur Chris Power, the company operates highly automated machining facilities ('factories') that produce precision components and assemblies primarily for aerospace and defense customers. The company uses robotics, AI, and software-defined manufacturing systems that targets bringing new production online in months rather than years - helping to replace a highly-fragmented series of thousands of family-owned, labor-intensive machining shops that contribute to the fragility of the A&D industrial base. Hadrian's model extends beyond parts production with a "Factories-as-a-Service" approach that integrates end-to-end processes from raw material to finished systems and short-cycle training to increase worker productivity. The company currently operates three major facilities in Torrance, CA and in Mesa, Arizona. The Mesa plant ('Factory 3') was [recently opened](#) in January, 2026 and spans more than 290,000 sq feet and will be used to make precision components for A&D applications.

On March 20th 2026, [Hadrian and the US Navy](#) announced the 2.2mn square foot Factory 4 (F4) in Cherokee, Alabama which would mass produce components for the Virginia-class attack submarines and Columbia-class ballistic missile submarines. It would be funded by more than \$2.4bn in investment consisting of \$900mn of funds from the One Big Beautiful Bill Act (OBBBA) and \$1.5bn in private capital.

According to PitchBook, the company has raised \$611mn to-date from notable investors including Founders Fund, a16z, Altimeter, Construct Capital, T. Rowe Price, Lux Capital, and Caffeinated Capital, among others. It was last valued at \$1.6bn as part of a \$391mn Series C round (PitchBook) in July 2025 and a later follow-on in January 2026.

Exhibit 28: Hadrian (Automated Machining/Manufacturing for A&D Applications)

HADRIAN

Torrance, CA



Hadrian Factory

Partners



ANDURIL



DIRAC



LOCKHEED MARTIN



U.S. NAVY

Notable Team Members



Chris Power
Founder & CEO
Prior:
ADGC Ventures
Ento.com



West Owens
CFO
Prior:
TerraWest
SolarCity



Lars Lider
VP Strategy
Prior:
Machine Sciences
KPMG

Notable Investors

\$611mm Funding Raised (\$1.6bn Valuation - Dec 2023)



FOUNDERS FUND



**andreessen
horowitz**



ALTIMETER



CONSTRUCT



T.RowePrice



**CAFFEINATED
CAPITAL**



Lux

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

Matter

Matter is a contract manufacturing startup that aims to build and operate fully autonomous factories using modular automation and its proprietary 'matterOS'. It was founded by Adi Prasad, Charly Mwangi, Aish Varadhan, and Aditya Ranjan - former manufacturing leaders at companies such as Tesla, Apple, Rivian, Intel, and Johnson Controls. The company claims its platform fully integrates design, planning, procurement, and production into a single software-defined workflow, with the goal of improving speed and flexibility as product cycles shorten and demand becomes less predictable. Matter reports that this approach relies on reusable automation and software orchestration to reduce manual labor and enable rapid changeovers between products, particularly for complex hardware such as robotics, energy systems, and consumer electronics. The company appears to remain in a largely stealth phase, and publicly available information on its operations, customers, and financials is limited.

Exhibit 29: Matter (Autonomous Factories for Contract Manufacturing)



matter

Menlo Park, CA

Founders



Charly Mwangi
Co-Founder, Exec Chairman
Prior:
Rivian
Tesla
Eclipse Ventures



Adi Prasad
Co-Founder, CEO
Prior:
Tesla
Apple
Rivian



Aish Varadhan
Co-Founder, CTO
Prior:
Rivian
Amazon
Apple
Intel



Matter Autonomous Contract Manufacturing Platform

Source: Company Websites, Morgan Stanley Research

VulcanForms

VulcanForms is a US-based digital manufacturing company focused on scalable metal additive production. The company was founded in 2015 out of the Massachusetts Institute of Technology with the aim of bringing additive manufacturing from prototyping into industrial scale production. The company grew out of collaboration between MIT Professor John Hart and Martin C. Feldmann, who met when Feldmann was a graduate student in Hart's advanced manufacturing course and later worked together on research in metal additive manufacturing and production systems. Their work at MIT focused on improving the speed, reliability, and repeatability of laser powder bed fusion (LPBF) processes, with an emphasis on making the technology suitable for high volume manufacturing environments. The two would eventually create a novel LPBF printer architecture that could run multiple lasers simultaneously (current machine has 75 lasers), boosting throughput without (per company claims) compromising finished component quality.

VulcanForms has moved from early pilot projects toward high-volume operations with a centralized factory model that coordinates machinery and data to compress lead times and increase throughput. Recent output includes test production of a 2 kN rocket thruster using its latest generation (Gen 3) additive system, illustrating technical progress with partners in aerospace supply chains.

According to PitchBook, the company has raised \$566mn to-date from notable investors including Eclipse Ventures, 1789 Capital, D1 Capital Partners, IEQ Capital, Fontinalis Partners, and Industry Ventures, among others.

See More: [VulcanForms: Metals Manufacturing, Made Simpler \(5 Feb 2026\)](#)

Exhibit 30: VulcanForms (Vertically Integrated 'Additive+' Metal Manufacturing)

VulcanForms™
Accelerating Innovation
Devens, MA

Notable Team Members

 Kevin Kasekert Founder & CEO Prior: Rodwood Materials – CDO Tesla	 John Hart Co-Founder, Board Member Prior/Owner: MIT – Prof. Mechanical Engineering	 Jay Martin President Prior: Globus Medical Intuitive Surgical
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Notable Investors
\$566mn Funding Raised (Valuation Unknown)

Vertically Integrated Additive & Subtractive Metals Manufacturing

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

Startups to Follow: Factory Brain

These are companies that provide key software or AI models that interconnect machines, data, and workers to manage, analyze, and optimize factory operations.

Arda

Arda is an early-stage AI startup co-founded by former OpenAI chief research officer Bob McGrew that is developing a software/AI platform to create autonomous factories. According to the [WSJ in March 2026](#), the company is developing an AI platform that uses video models to analyze factory floor operations and train robots to perform tasks autonomously, while coordinating both machines and human workers across multiple stages of the production process (McGrew has not publicly commented to our knowledge). Arda is also reportedly in the process of raising about \$70 million at an approximately \$700 million valuation in a seed round led by Founders Fund and Accel, with participation from Khosla Ventures and XYZ Venture Capital (though reporting notes that terms are subject to change). We note that Arda is currently operating in stealth, and limited technical or product details have been disclosed to-date.

Exhibit 31: Arda (Stealth Bob McGrew Autonomous Factory Startup)

Arda

(In Stealth)

Founder



Bob McGrew
Prior:
Chief Research Officer – OpenAI
Palantir
PayPal

- Reportedly building a software platform for autonomous factories.
- Platform to coordinate machines and humans from product design to finished goods.
- Also creating a video model to analyze factory floors to train autonomous robots.
- Reportedly raising \$70mn at a \$700mn valuation led by Founders Fund and Accel

Morgan Stanley

Source: Company Websites, The Wall Street Journal, Morgan Stanley Research

Atoms

Atoms is a robotics and automation company led by Uber co-founder and former CEO Travis Kalanick that is developing a standardized robotics platforms, software, and infrastructure for a range of physical industries. Initial focuses appear to be on sectors such as food service, mining, and transportation. The company exited stealth in March 2026 and rolled up a group of businesses including [CloudKitchens](#) (ghost kitchen and food delivery infrastructure), [Pronto AI](#) (Autonomous mine/quarry haulage), [Lab 37](#) (automated food assembly), [Otter](#) (restaurant software OS), [Picnic](#) (office food delivery), and [ProFood](#) (food production real estate manager), with plans to add additional companies over time -

including one in the transport industry. Atoms aims to deploy what it calls “gainfully employed robots,” or specialized machines designed to perform specific tasks at industrial scale with one of the products described as a modular wheel base for robotics. Details on commercial deployment and technical performance remain limited following its public launch.

While info is sparse at this stage, [job postings](#) from the company provide detail to the company's future plans: *"From food production and delivery to mining and transport, ATOMS develops robotics, software, and infrastructure that automate how things are produced and moved through the physical world. Our systems combine robotics and intelligent software to sense, predict, and act in real-world environments, making earth's essential industries more productive, reliable, and scalable."*

Exhibit 32: Atoms (Automation Platforms for Food Production, Delivery, Mining, Transport)

ATOMS

Founder

Travis Kalanick
Founder, CEO
Prior:
Co-Founder/CEO Uber

- Standardized robotics platforms, software, and infrastructure for a range of physical industries including: Delivery, Food Production, Mining, Transport
- Exited Stealth in March 2026. Official products/robots yet to be unveiled

Companies

CloudKitchens® PRONTO
Otter Picnic
Lab37 ProFood Properties

Morgan Stanley

Source: Company Websites, Morgan Stanley Research

Intrinsic (Part of Alphabet - Covered by Brian Nowak)

Intrinsic is a robotics software company within Alphabet that develops AI-based tools to make industrial robots easier to program and deploy across manufacturing and logistics. The company began inside Alphabet's X division and launched as a standalone unit in 2021, focusing on combining machine learning, computer vision, and motion planning into a unified software platform (the 'Intrinsic Platform') that reduces integration time and cost. Intrinsic does not build hardware. It provides a software layer that works across robots from multiple manufacturers and aims to function as an 'Android for robots,' giving developers a common environment to build, run, and fine-tune applications across different systems. Alphabet recently folded Intrinsic into Google's main operations (from the "Other Bets" category), working closely with DeepMind and its Gemini Robotics models, with a focus on scaling physical AI in real-world industrial settings. The company is led by CEO Wendy Tan-White, a former Alphabet X executive and co-founder of Moonfruit, who has overseen Intrinsic since its 2021 launch and continues to guide its product and commercial strategy.

Exhibit 33: Intrinsic (Industrial Robot Design and Deployment Platform)

The slide features the 'intrinsic' logo at the top left, with 'Mountain View, CA' below it. The central graphic shows a 3D CAD model of a robotic arm in a factory setting. Below this, a diagram illustrates the 'Intrinsic solution lifecycle' with stages: Design → Develop → Validate → Install → Operate → Maintain → Adjust. It also shows a flow from 'Developer' (User: Powersolution build and install solution) to 'Operator' (User: app-specific (HMI) to operate solution), with 'intrinsic' in the middle. The bottom text reads 'The Intrinsic Platform' and 'Support for Robotic Systems from Development through Production'.

Notable Team Members

 Wendy Tan-White CEO Ex: BGF Ventures, Moonfruit	 Torsten Kroeger Chief Science Officer Ex: Reflexion GmbH, Loom Vision GmbH	 Jake McCarter Head of Platform Product Ex: Viomi, Boss Nova Robotics, L.E.K. Consulting
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Notable Partners

 鴻海科技集團	

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

Mind Robotics

Mind Robotics, founded in 2025 by Rivian CEO/Founder RJ Scaringe, is an industrial robotics startup that aims to develop AI-enabled robots for manufacturing and factory environments. The company has reportedly raised \$615 million to date, including a recent \$500 million Series A round (co-led by Accel and Andreessen Horowitz) that valued the company at about \$2 billion. Mind Robotics states that it focuses on automating complex, variable tasks that traditional industrial robots struggle to handle, such as work requiring dexterity, adaptation, and real-time decision-making. The company has disclosed limited technical detail, but indicates it is building an integrated system that combines robotic hardware with AI foundation models trained on real-world production data as well as key deployment infrastructure. It maintains a strategic partnership with Rivian, which provides production data and initial deployment sites. The founding team includes engineers from Physical Intelligence, Waymo, Zoox, and Google, with experience in autonomy and AI systems. The company has not disclosed detailed product specifications, timelines, or broader commercialization plans.

Exhibit 34: Mind Robotics (Generalizable Robotics for Hard-to-Automate Factory Tasks)

MIND ROBOTICS

Founding Team

Team From:
Physical Intelligence
Waymo
Zoox
Google

Notable Investors
\$615mn Funding Raised (~\$2.0 Valuation - March 2026)

Accel **ECLIPSE**
andreessen horowitz

- Building intelligent robots for hard-to-automate factory tasks
- Official products/robots yet to be unveiled
- Strategic partnership with Rivian for robot deployments + production scale data harvesting

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

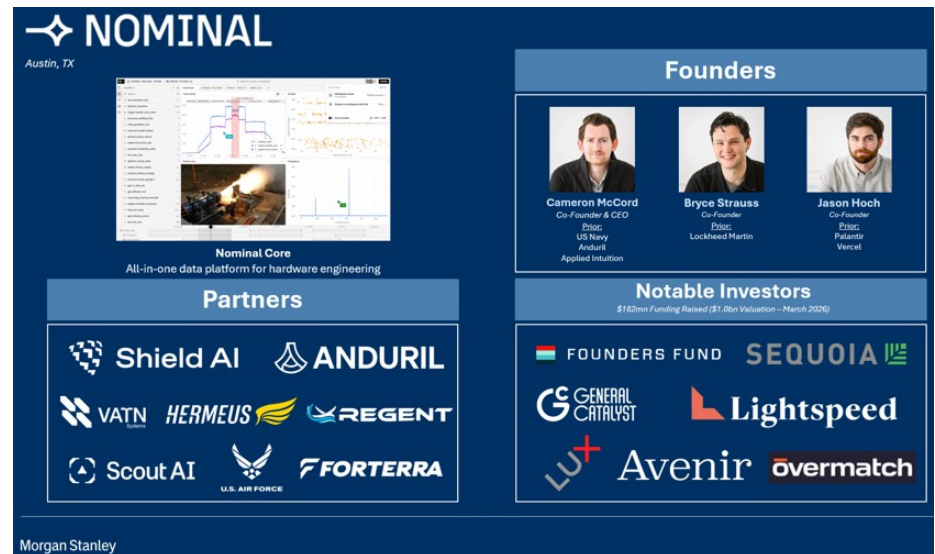
Nominal

Nominal is a US-based software startup building a unified data and testing platform for engineering teams developing complex hardware systems. The company was founded in Austin, TX in 2022 by Cameron McCord, Jason Hoch, and Bryce Strauss and provides a real-time software stack that enables engineers to collect, analyze, and monitor telemetry and operational data across the lifecycle of physical systems, from early prototyping and lab testing to live operational deployments. Nominal's platform (including products such as **Nominal Core** and **Nominal Connect**) aggregates data from sensors, test equipment, simulations, and logs into a single collaborative workspace, allowing teams to automate hardware-in-the-loop testing, monitor system performance in real time, and rapidly analyze results. The software is designed to address a major bottleneck in modern hardware development, where engineering teams often rely on fragmented scripts, spreadsheets, and legacy tooling to process large volumes of testing data.

Nominal's platform is used across aerospace, defense, robotics, energy, and advanced manufacturing by organizations developing critical systems such as satellites, autonomous aircraft, drones, and hypersonic technologies. The company works with partners and customers including **Anduril**, **Shield AI**, **the US Air Force**, **Hermeus**, **Forterra**, and many others. These organizations use the platform to analyze telemetry, surface anomalies, and accelerate testing cycles for advanced hardware programs.

According to PitchBook, the company has raised \$182mn to-date from notable investors including Sequoia Capital, Founders Fund, General Catalyst, Lux Capital, Lightspeed Venture Partners, among others. It was last valued at \$1.0bn in March 2026 as part of a **\$155mn Series B round**.

Exhibit 35: Nominal (Unified Industrial Data Stack for Testing & Validating Complex Hardware Systems)



Source: Company Websites, PitchBook, Morgan Stanley Research

Project Prometheus

Project Prometheus is a recently launched American industrial AI startup co-founded by Jeff Bezos and **Vik Bajaj** (who serve as co-CEO's) in 2025, marking Bezos's first formal executive role since he stepped down as Amazon CEO. The company (currently in stealth) is building AI systems to address engineering, robotics, and manufacturing challenges in sectors such as aerospace, automobiles, and computing by enabling models that learn from real-world physical data - according to [external reporting](#). Per the NYT in November, it had already recruited a team of over 100 researchers from leading AI labs including OpenAI, DeepMind, and Meta and quietly acquired the agentic AI firm General Agents in June to bolster its technology stack. Early public details on Project Prometheus's headquarters, specific product roadmap, and customer base remain limited, but its focus on "AI for the physical economy" highlights ambitions to integrate advanced AI into industrial processes and complex hardware workflows.

According to PitchBook, the company has raised \$6.2bn (valuation unknown) - partly from Jeff Bezos, with the remainder from undisclosed investors.

Separately, it's recently been reported by the WSJ that Bezos is in the process of raising **\$100bn for an AI manufacturing fund** which is separate from but related to Project Prometheus (Jeff Bezos has not commented to our knowledge). The fund would reportedly buy up manufacturing companies in industries such as chipmaking, defense, and aerospace to integrate them with AI. If completed, it would be the largest single fund raise for an investment fund since the \$100bn launch of SoftBank Vision Fund in 2017.

Exhibit 36: Project Prometheus (Stealth Bezos Manufacturing AI Startup)

Project Prometheus
(In Stealth)

Co-Founders & Co-CEO's



Jeff Bezos
Prior:
Founder & CEO Amazon
Founder Blue Origin



Vik Bajaj
Prior:
CEO Foresite Labs
Co-Founder Google Life Sciences
Chief Science Officer Verily

- Launched 2025 to build AI for engineering and manufacturing
- Official products yet to be unveiled
- 100+ Employees from OpenAI, Google DeepMind, Meta, and others.
- \$6.2bn in Funding (PitchBook) – partly from Jeff Bezos.

Morgan Stanley

Source: Company Websites, New York Times, PitchBook, Morgan Stanley Research

Tulip Interfaces

Tulip Interfaces is a US-based industrial software startup developing a frontline operations platform that enables manufacturers to digitize and optimize shop-floor processes. The company was founded out of the MIT Media Lab by Natan Linder and Rony Kubat and is headquartered in Somerville, MA. Tulip provides a cloud-based, no-code platform that allows factory teams to build simple applications that guide workers through production steps, track quality checks, and collect data from machines and operators on the shop floor. By connecting workers, equipment, and production data in a single system, Tulip helps manufacturers replace paper instructions, spreadsheets, and disconnected software with digital workflows that are easier to update and monitor in real time.

Tulip's platform is used across manufacturing industries including pharmaceuticals, aerospace, medical devices, electronics, and industrial equipment. The company works with partners and customers including DMG MORI, Sartorius, Stanley Black & Decker, Johnson & Johnson, and Vertex Pharmaceuticals, among others. These organizations use Tulip's software to guide operators through complex assembly or production processes, track issues as they occur, and gain better visibility into how factories are performing.

According to PitchBook, the company has raised \$273mn to-date from notable investors including Mitsubishi Electric, Breyer Capital, Insight Partners, and New Enterprise Associates, among others. It was last valued at \$1.3bn in a \$120mn Series D round led by Mitsubishi Electric in December 2025.

Exhibit 37: Tulip Interfaces (Frontline Factory Operations Management)


TULIP
Somerville, MA

Tulip Frontline Operations Platform

Founders

Rony Kubat
Co-Founder, CIO, CISO
Bior
MIT Media Lab

Natan Linder
Co-Founder, CEO
Elior
Formlabs
MIT Media Lab
Samsung Telecom, Research Israel

Notable Partners

Johnson & Johnson
DMG MORI
SARTORIUS
VERTEX
MITSUBISHI ELECTRIC
Stanley Black & Decker

Notable Investors
\$275mn Funding Raised (\$1.2bn Valuation - Dec, 2023)

MITSUBISHI ELECTRIC
breyer capital
vertex VENTURES
INSIGHT PARTNERS
NEA
Stanley Black & Decker

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

Vention

Vention is a Canada-based manufacturing technology company that develops a cloud-enabled platform for designing and deploying robots and other automation equipment. The company was founded in Montreal in 2016 by Etienne Lacroix and Max Windisch and provides software that allows manufacturers to design, simulate, and build custom automation systems such as robotic work cells, assembly lines, and material handling equipment. Through Vention's interface, engineers can configure factory equipment using a library of modular hardware components, create digital models, and order the required parts through an integrated supply chain. The platform helps companies build and deploy automation systems more quickly and with less custom engineering than traditional factory automation projects.

The company also offers a marketplace of standardized hardware components including frames, motion systems, conveyors, and robot integrations that can be combined to create automated production setups. Partnerships with companies such as [ABB](#), [Universal Robots](#), [FANUC](#), [Franka Robotics](#), [Epson](#), [Doosan](#) and [NVIDIA](#) allow Vention's platform to integrate industrial robots and advanced computing capabilities. This combination of software, modular hardware, and technology partners is intended to make factory automation easier to design, purchase, and deploy for manufacturers.

According to PitchBook, the company has raised \$263mn to-date (last raised \$110mn in a [Series D funding round in January 2026](#)) from investors including NVentures, Bain Capital Ventures, Fidelity Investments Canada ULC, Investissement Québec, and Desjardins Capital.



VENTION

Montreal, Canada

Overview



Vention AI-Powered Automation Platform

Notable Partners



UNIVERSAL
ROBOTS



FANUC



EPSON



DOOSAN



ABB



NVIDIA



FRANKA ROBOTICS

Founders



Etienne Lacroix
Co-Founder, CEO
Prior:
McKinsey
General Electric



Max Windisch
Co-Founder, CSO
Prior:
Microsoft
EMC
PI

Notable Investors

\$265mm Funding Raised (Valuation Unknown)



NVIDIA



BainCapital
VENTURES



Fidelity
INVESTMENTS



Investissement
Québec



Desjardins
Capital

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

Startups to Follow: General-Purpose Robot Foundation Models

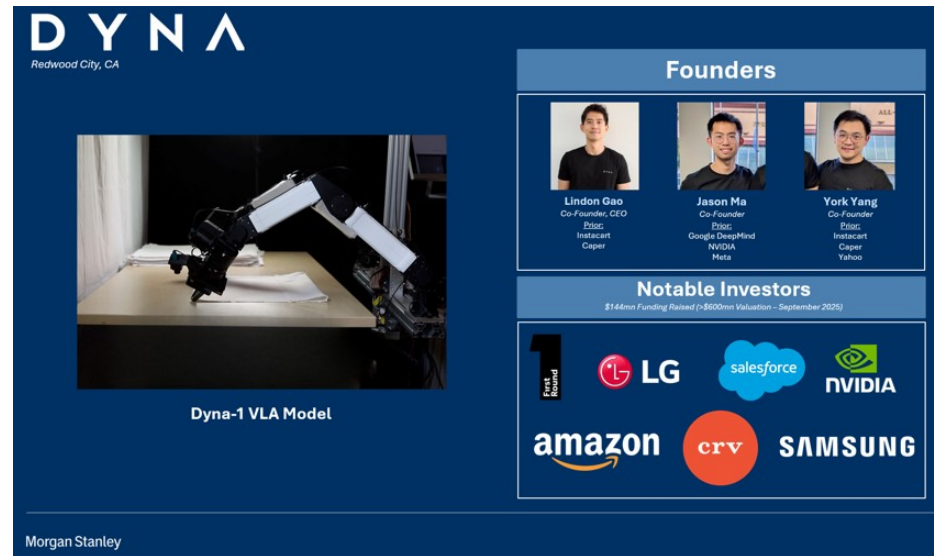
These are companies building AI models that aim to control many types of robots and enable them to perform a wide range of physical tasks. They train on large datasets from simulations and real-world environments with the goal of reducing the need for task-specific programming by enabling robots to generalize across use cases.

Dyna

Dyna Robotics is an AI and robotics company developing general-purpose robotic systems for real-world applications. Established in 2024 and based in California, the company was started by Lindon Gao, a repeat founder who previously co-founded Caper AI, which sold to Instacart for about \$350 million.

The company builds robotics foundation models that integrate perception and control into a single system. Its current model, [DYNA-1](#), powers dual-arm robots deployed in commercial settings such as restaurants, laundromats, and factories. These systems perform tasks such as folding, food preparation, and packaging, with the company [claiming it has achieved an over 99% task success rate](#) in a specific continuous 24-hour operation where it folded 850+ napkins autonomously with zero interventions. Dyna Robotics claims its models improve with real-world use and can adapt to new environments with limited additional training. The company targets industries including food service, manufacturing, and other labor-intensive service operations.

According to PitchBook, the company has raised \$144mn to-date, most recently in September 2025 when it raised a \$120mn [Series A round](#) led by First Round Capital, CRV, and RoboStrategy with participation from Salesforce Ventures, NVentures, Amazon Industrial Innovation Fund, Samsung NEXT Ventures, LG Technology Ventures, and others - valuing it at >\$600mn.

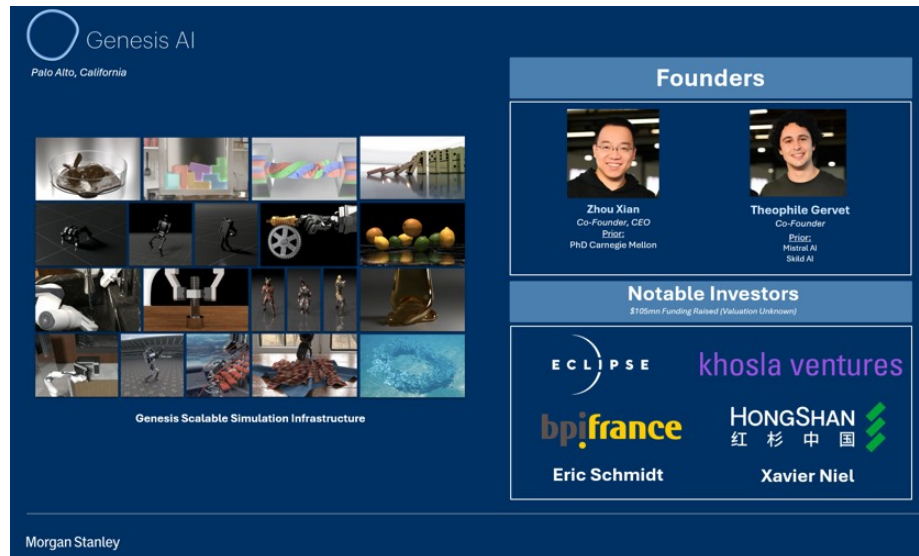
Exhibit 39: Dyna (General Purpose Robots & VLA Foundation Models)

Source: Company Websites, PitchBook, Morgan Stanley Research

Genesis

Genesis AI is a US-based full-stack robotics lab that is developing a universal robot foundation model and simulated data engine. In simple terms, the company is building AI-powered systems that allow robots to handle many different physical tasks instead of being limited to a single job. It is currently based in Palo Alto, and **emerged from stealth in July 2025 with \$105mn in funding co-led by Khosla Ventures and Eclipse Ventures.** It is founded by researchers spanning notable AI labs including Mistral AI, Nvidia, and Google and academics from CMU, MIT, Columbia, and UMD. The company is currently led by **CEO Zhou Xian**- a CMU PhD who initiated the original Genesis project- a physics platform designed for general-purpose physical AI.

Its approach is built around three core pillars: 1) Genesis is developing a scalable data engine that combines real-world robot interactions, high-quality physics simulations, and large-scale embodied data from the internet to train a universal robotics model capable of controlling many types of robots across different tasks; 2) the company is fostering an open-source ecosystem, **sharing simulation tools and model weights** to accelerate progress across the broader robotics community; and 3) Genesis focuses on real-world deployments, testing and refining its systems in messy, unstructured environments to ensure robots can operate reliably outside of controlled settings.

Exhibit 40: Genesis AI (General Purpose Foundation Model & Data Engine for Robots)

Source: Company Websites, PitchBook, Morgan Stanley Research

Physical Intelligence

Physical Intelligence is an AI and robotics company building general-purpose VLA Models for a wide range of robot systems, trained predominately on diverse real-world data. Founded in 2024 and based in San Francisco, the company brings together researchers from leading institutions such as Google DeepMind & Brain, Stanford, and UC Berkeley to develop foundation models that allows robots to operate more flexibly across environments.

The company builds vision-language-action (VLA) models for robotics that integrate perception, reasoning, and control into a single system. These models interpret visual inputs, process natural language instructions, and translate them into physical actions, enabling robots to generalize across tasks without requiring reprogramming for each new use case. Early demonstrations show systems folding laundry and performing basic manipulation tasks in unstructured environments - including making coffee, cleaning rooms, etc. According to the company's leadership, these models are trained on real-world data collected from diverse operating environments with simulation secondary.

Physical Intelligence's recent research has explored reinforcement learning (RL) approaches that allow robots to improve through real-world experience. The company describes combining offline training with on-robot data collection, where systems learn from both autonomous actions and human feedback to refine performance over time, with early results showing gains in task efficiency and reliability.

According to PitchBook, the company has raised \$1.07bn to-date from notable investors including Sequoia Capital, CapitalG, Thrive Capital, Lux Capital, Alkeon Capital Management, NVentures, T. Rowe Price, and Open AI, among others. It was last valued at \$5.6bn in a \$600mn Series B funding round in November 2025.

Exhibit 41: Physical Intelligence (General Purpose Robot Foundation Model)

Physical Intelligence (π)
San Francisco, CA

$\pi^*_{0.6}$ VLA Model

Notable Investors
\$1.67bn Funding Raised (\$5.6bn Valuation – November 2025)

khosla ventures, OpenAI, BEZOS EXPEDITIONS, LUT, THRIVE CAPITAL, SEQUOIA, CapitalG

Key Founders

 Karol Hausman Co-Founder, CEO Prior: Google DeepMind	 Sergey Levine Co-Founder Prior: UC Berkeley Professor	 Chelsea Finn Co-Founder Prior: Stanford Professor, Google DeepMind	 Brian Ichter Co-Founder Prior: Google DeepMind
 Lachy Groom Co-Founder Prior: Stripe	 Quan Vuong Co-Founder Prior: Google DeepMind	 Suraj Nair Co-Founder Prior: Toyota Research Institute	

Morgan Stanley

Source: Company Websites, New York Times, PitchBook, Morgan Stanley Research

Skild AI

Skild AI is a robotics and artificial intelligence company developing a general-purpose “robot brain” pre-trained heavily on human video and simulation. The company trains its models primarily on large-scale internet video data/[human videos](#) and simulation to help machines learn from real-world human activity and operate in unstructured environments. Skild AI was founded in 2023 and is headquartered in Pittsburgh by Deepak Pathak and Abhinav Gupta, both current professors at Carnegie Mellon’s Robotics Institute with research in robotic dexterity, world models, and learning from human video.

The company develops a robotics foundation model called the ‘[Skild Brain](#)’, which it describes as a scalable AI system for controlling a wide range of robots. Traditional robot software is typically built for a specific machine and narrowly defined tasks in controlled environments. Skild AI aims to move beyond this approach by training a single model that can adapt across different robot form factors, tasks, and settings. Leveraging both simulation and internet-scale data, the company claims it trains on at least **1,000x** more data points than competing approaches. Example capabilities include object manipulation, navigation, and task execution across platforms such as robotic arms, mobile robots, and humanoids.

According to PitchBook, the company has raised \$2.21bn to-date from notable investors including SoftBank, Nvidia, Samsung, LG, Coatue, Sequoia, In-Q-Tel, Bezos Expeditions, and others. It was last valued at **>\$14.0bn in a \$1.4bn Series C funding round** in January 2026.

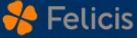
Exhibit 42: Skild AI (General Purpose Robot Foundation Model Trained on Human Video + Simulation)

SKILD AI
Pittsburgh, PA

Founders

 Deepak Pathak, PhD Co-Founder, CEO Other Titles: Professor - Carnegie Mellon Facebook	 Abhinav Gupta, PhD Co-Founder, President Other Titles: Professor - Carnegie Mellon Facebook Google
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Notable Investors
\$2.21bn Funding Raised (+\$14.0 Valuation - January 2020)

   	   
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'Skild Brain'
General-Purpose Robotics Foundation Model

Morgan Stanley

Source: Company Websites, PitchBook, Morgan Stanley Research

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Tesla Inc (TSLA.O)

Our PT of \$415 is comprised of 5 components: **(1)** \$45/share for core Tesla Auto business on 8.5mm units in 2040, 9.8% exit EBIT margin (ex-FSD), 10.9% WACC, and 10x 2030 exit EBITDA multiple. **(2)** Network Services at \$145/share, 80% attach rate at \$240/month ARPU by 2040. **(3)** Tesla Mobility at \$125/share on DCF with ~5mn cars at ~\$1.33/mile by 2040. **(4)** Energy at \$40/share, and **(5)** Humanoids at \$60/share (50% probability discount).

Risks to Upside

- Increased FSD attach rate / unsupervised FSD
- Cost milestones on new battery
- New model intro (Roadster, multivan)
- Robotaxi rollout
- Optimus announcement

Risks to Downside

- Competition: legacy OEMs/Chinese players/big tech in Autos and Robotics
- Execution risk across robotaxi/FSD/humanoid
- Regulatory
- China risk
- Dilution
- Valuation

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(as of February 28, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1528	41%	444	49%	29%	690	42%
Equal-weight/Hold	1590	43%	374	41%	24%	725	44%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	588	16%	87	10%	15%	220	13%
Total	3,710		906			1636	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

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Stock Price, Price Target and Rating History (See Rating Definitions)

Tesla Inc (TSLA.0) - As of 03/26/26 GMT in USD
Industry : Autos & Shared Mobility



Stock Rating History: 3/1/21 : O/I; 6/22/23 : E/I; 9/10/23 : O/I; 3/13/24 : O/A; 9/25/24 : O/I; 12/7/25 : E/I

Price Target History: 2/1/21 : 293.33; 4/22/21 : 300; 10/24/21 : 400; 1/10/22 : 433.33; 6/22/22 : 400; 7/14/22 : 383.33; 8/25/22 : 383; 10/10/22 : 350; 10/24/22 : 330; 12/28/22 : 250; 1/25/23 : 220; 4/20/23 : 200; 6/22/23 : 250; 9/10/23 : 400; 10/19/23 : 380; 1/22/24 : 345; 3/5/24 : 320; 4/4/24 : 310; 12/10/24 : 400; 1/13/25 : 430; 3/20/25 : 410; 12/7/25 : 425; 1/29/26 : 415

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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